

Adaptive Post-Quantum Secure UAV Communication

Under Energy and Adversarial Constraints

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Abstract—Small unmanned aerial vehicles communicate over plaintext MAVLink links that face both classical and emerging quantum threats. We present a transparent post-quantum cryptographic proxy for a Raspberry Pi 4 companion computer that protects MAVLink traffic without firmware modification and adapts its security posture at runtime. The system spans 72 cipher suites—nine key encapsulation mechanisms, eight digital signatures, and three authenticated ciphers across NIST security levels 1, 3, and 5—whose handshake costs range from 1.5 ms to over 11 s. At the core of the design is the *Multi-Dimensional Energy-Aware Scheduler* (MDEAS), a 15-step reactive policy cascade that jointly monitors battery voltage, processor temperature, and link quality to autonomously select the strongest cipher suite sustainable under current operating conditions. Using an INA219 current sensor at 1 kHz, we measure per-operation power and energy for every primitive, quantify the combined overhead of post-quantum cryptography and on-board DDoS detection, and show that MDEAS maintains uninterrupted telemetry even when concurrent threat detection raises system power by 59%.

Index Terms—Post-quantum cryptography, UAV communication, adaptive security, energy-aware scheduling, MAVLink, embedded systems

I. INTRODUCTION

A growing share of civil and commercial drone operations—precision agriculture, infrastructure inspection, search and rescue—depend on continuous telemetry between an airborne autopilot and a ground control station. The dominant protocol for this link, MAVLink version 2 [1], was engineered for bandwidth-constrained radio channels. It carries heartbeats, GPS fixes, and actuator commands in a compact binary format, yet it provides no encryption: an optional 48-bit signature offers integrity but every byte of payload remains in plaintext.

This openness creates two distinct threat horizons. In the near term, a passive eavesdropper with a software-defined radio can reconstruct a complete mission profile—waypoints, altitude, battery state—from intercepted telemetry. In the longer term, a cryptanalytically relevant quantum computer will render the elliptic-curve and RSA primitives that underpin today’s key exchange vulnerable to Shor’s algorithm [2]. The *harvest-now, decrypt-later* strategy makes even current recordings a future liability when captured telemetry carries a decade-long operational shelf life.

The natural response is to deploy post-quantum cryptography (PQC). NIST concluded its standardisation in 2024, publishing ML-KEM (FIPS 203) [3] for key encapsulation and ML-DSA (FIPS 204) [4] for digital signatures, with hash-based SLH-DSA (FIPS 205) [5] and code-based candidates

HQC [6] and Classic McEliece [7] still under evaluation. Yet a drone is not a data centre. Its companion computer runs on a shared power rail, contends with propulsion heat, and must sustain real-time telemetry at 5–50 Hz. A single heavyweight handshake—Classic McEliece at NIST Level 5 takes 9 s and draws 40 J on a Cortex-A72—can stall the control link and drain a measurable fraction of the flight battery. Choosing a fixed cipher suite is therefore inadequate: the right algorithm at takeoff may be the wrong one during a thermal-constrained aggressive manoeuvre ten minutes later.

This observation motivates our core contribution. Rather than benchmarking PQC primitives in isolation, we treat suite selection as a *runtime decision* governed by the aircraft’s instantaneous energy, thermal, and link-quality state. We introduce the **Multi-Dimensional Energy-Aware Scheduler** (MDEAS), a 15-step reactive policy cascade that monitors three telemetry axes and autonomously navigates a 72-suite design space, downgrading security level or shedding on-board DDoS detection when operating constraints demand it and restoring full protection when conditions permit.

In summary, this paper makes four contributions:

- 1) A **transparent PQC proxy** for MAVLink traffic that performs a two-message handshake authenticated by a post-quantum signature and encrypts every subsequent packet with authenticated encryption, requiring no autopilot firmware changes.
- 2) **Systematic embedded benchmarks** of 20 post-quantum algorithms (9 KEMs, 8 signatures, 3 AEADs) on a Raspberry Pi 4 at a locked 1.8 GHz, capturing per-operation execution time, power draw, and energy through an INA219 current sensor at 1 kHz—the first such measurement suite on ARM Cortex-A72 hardware with component-level power instrumentation.
- 3) **The MDEAS scheduler**, a 15-step reactive policy cascade that jointly monitors battery voltage, processor temperature, and link quality to select the strongest feasible cipher suite at runtime, with explicit cross-axis exclusion rules and detector-aware thermal budgeting.
- 4) **Quantification of the PQC-DDoS interaction**: we show that on-board XGBoost detection adds 951 mW (29% of baseline) while a Transformer model doubles CPU utilisation to 93%, and demonstrate that MDEAS absorbs this overhead without telemetry loss.

The remainder of this paper is organised as follows. Section II presents the system architecture. Section III introduces

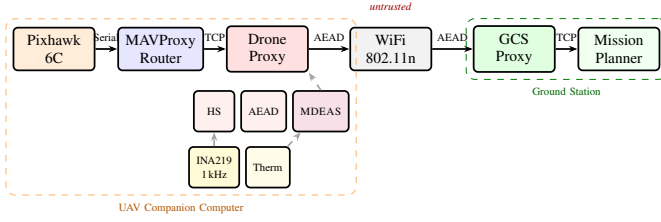


Fig. 1. Deployment architecture. The cryptographic proxy runs alongside MAVProxy on the RPi 4 companion computer. MDEAS receives telemetry from INA219 and thermal sensors and adjusts the AEAD engine and handshake module at runtime. No firmware modification is required on the Pixhawk autopilot.

the adaptive security model. Section IV describes the experimental setup. Section V reports results. Section VI discusses related work, and Section VII concludes.

II. SYSTEM ARCHITECTURE

The architecture, shown in Fig. 1, interposes a transparent proxy between the autopilot and the wireless link. On the drone side, MAVProxy forwards MAVLink packets over a local TCP socket to the Drone Proxy, which performs a PQC handshake with its ground-side counterpart. After key agreement, every packet is wrapped in an authenticated-encryption envelope before crossing the WiFi 802.11n channel. The ground proxy decrypts and delivers plaintext MAVLink to the mission planner.

Three communication planes share this tunnel:

- **Control plane:** heartbeats, mode changes, command acknowledgements (5–10 Hz, 64–128 B).
- **Telemetry plane:** GPS, attitude, battery voltage (10–50 Hz, 128–512 B).
- **Bulk plane:** parameter blocks, log downloads, firmware updates (burst, 1–4 kB).

The proxy design is intentionally transparent: the Pixhawk runs unmodified ArduPilot firmware [8], and the mission planner sees a standard MAVLink connection. This separation allows cryptographic upgrades—new algorithms, policy updates—without reflashing the flight controller, a practical necessity for fleets in the field.

A cipher suite in our system combines one key encapsulation mechanism (KEM), one digital signature (SIG), and one authenticated cipher (AEAD), constrained to the same NIST security level. Nine KEMs span three families (lattice: ML-KEM; code-based: HQC, Classic McEliece), eight signatures span three families (lattice: ML-DSA; NTRU-lattice: Falcon; hash-based: SPHINCS⁺), and three AEADs (AES-256-GCM, ChaCha20-Poly1305, Ascon-128a) complete the data plane. Level-matching yields $27 + 18 + 27 = 72$ suites across L1, L3, and L5 (Falcon lacks an L3 variant, leaving only two signatures at that level).

III. ADAPTIVE SECURITY MODEL

A fixed cipher suite is inappropriate for a platform whose thermal envelope, battery state, and link quality shift contin-

uously through a mission. ML-KEM-768 completes a handshake in 2.7 ms, consuming 10.7 mJ, while Classic McEliece-8192128 needs 9.2 s and 40.6 J—a gap exceeding three orders of magnitude. If the scheduler blindly selects the strongest available suite, a single rekey during a thermal-constrained hover can trigger CPU throttling and degrade all three communication planes simultaneously.

MDEAS resolves this by treating suite selection as a constrained optimisation over three observable axes: **battery voltage**, **processor temperature**, and **link quality**. Formally, let $\mathbf{x}(t) = (V_{\text{bat}}, T_{\text{CPU}}, q_{\text{link}})$ denote the observable state at time t , and let $\mathcal{S}_f(\mathbf{x}) \subseteq \mathcal{S}$ be the subset of 72 suites satisfying all resource and exclusion constraints at $\mathbf{x}$. At each one-second scheduling epoch, MDEAS selects the highest-security-level suite in $\mathcal{S}_f$, preferring stronger AEAD and active DDoS detection when the thermal and energy budgets permit. Each axis is partitioned into operating regions with hysteretic thresholds to prevent oscillation between adjacent states.

A. Telemetry Acquisition

The drone’s local monitor samples three streams at 1 Hz, aggregated over a 5-second sliding window:

- **Battery:** INA219 voltage via I²C at 1 kHz. Three thresholds partition the range: *warning* (≤ 15.2 V), *low* (≤ 14.8 V), *critical* (≤ 14.0 V).
- **Temperature:** CPU thermal zone via `/sys/class/thermal`. *Warning* at 70 °C, *critical* at 80 °C.
- **Link quality:** Packets-per-second (floor: 5.0 pps) and maximum inter-packet gap (ceiling: 1000 ms).

B. Three-Axis Decision Space

MDEAS navigates three independent axes at runtime:

- **Axis 1 — AEAD cipher:** select Ascon-128a (lowest latency), ChaCha20-Poly1305, or AES-256-GCM based on per-packet timing budget.
- **Axis 2 — Security level:** move between L5, L3, and L1, trading handshake energy (from 12.6 mJ at K5+D5 down to 5.4 mJ at K1+F1) against NIST security margin.
- **Axis 3 — DDoS detector:** shed the Transformer (TST, +1.97 W, +10.7 °C) to XGBoost (+0.95 W, +4.8 °C) or disable detection entirely when CPU or thermal budgets are exhausted.

Two cross-axis combinations are explicitly forbidden. First, Ascon-128a is barred when the TST detector is active because TST’s sustained 93 % CPU utilisation starves Ascon’s software-only cipher path. Second, Classic McEliece paired with SPHINCS⁺ is prohibited because their combined handshake exceeds the 45-second timeout guard—a constraint validated by three benchmark failures at the McEliece-460896 + SPHINCS⁺-192s corner.

C. Policy Cascade

Algorithm 1 formalises the 15-step decision sequence evaluated every one second. Steps are ordered by severity: safety-critical conditions (stale telemetry, active DDoS, battery

Algorithm 1 MDEAS Policy Cascade (1 s period)

```
1: if telemetry stale ( $>5$  s) then
2:   hold current suite ▷ no data, no action
3: else if DDoS detector fires then
4:   switch to fastest AEAD ▷ minimise per-pkt cost
5: else if battery  $\leq 14.0$  V (critical) then
6:   downgrade to L1 + lightest suite
7: else if  $T_{\text{CPU}} \geq 80^\circ\text{C}$  (critical) then
8:   downgrade AEAD; shed detector if running
9: else if  $T_{\text{CPU}} > T_{\text{max}}^{\text{det}}$  then
10:  shed DDoS detector ▷  $75^\circ\text{C}$  XG,  $69^\circ\text{C}$  TS
11: else if  $T_{\text{CPU}} > 0.9 \times T_{\text{crit}}$  then
12:  preemptive AEAD downgrade ▷ predictive
13: else if AEAD latency exceeds budget then
14:  switch AEAD (Ascon  $\rightarrow$  ChaCha20)
15: else if AEAD latency below recovery then
16:  restore preferred AEAD
17: else if link degraded (pps  $< 5$  or gap  $> 1$  s) then
18:  hold; suppress rekey
19: else if cooldown active then
20:  hold ▷ stabilise after recent switch
21: else if proactive rekey interval elapsed then
22:  trigger two-phase commit rekey
23: else if detector under stress then
24:  downgrade detector (TS  $\rightarrow$  XG  $\rightarrow$  off)
25: else if security level promotable then
26:  upgrade KEM/SIG to next level
27: else
28:  hold: nominal operation
29: end if
```

critical, thermal critical) take precedence over optimisation-oriented actions (proactive rekey, detector upgrade, security-level promotion). Asymmetric hysteresis prevents oscillation: downgrade thresholds are set tighter than upgrade thresholds, so the system degrades quickly but restores cautiously.

D. Detector-Aware Thermal Model

MDEAS maintains a per-detector overhead model. XG-Boost adds 0.95 W and 4.8°C above baseline; TST adds 1.97 W and 10.7°C . When a detector’s thermal contribution would push T_{CPU} past its per-detector ceiling (75°C for XG-Boost, 69°C for TST), the cascade sheds the detector *before* downgrading the cipher suite—preserving cryptographic confidentiality at the expense of anomaly detection. This ordering reflects the judgement that intercepted cleartext is a permanent information loss, whereas a brief detector gap can be tolerated if the link remains encrypted.

IV. EXPERIMENTAL SETUP

A. Testbed

The UAV-side proxy runs on a Raspberry Pi 4 Model B Rev 1.5 (Broadcom BCM2711, quad-core Cortex-A72 at 1.8 GHz, 4 GB LPDDR4, Debian 12, kernel 6.12). The CPU governor is locked to `performance` to eliminate frequency-scaling artefacts. A Holybro Pixhawk 6C mini running ArduPilot [8] generates continuous heartbeat traffic over USB serial at 57 600 baud.

An INA219 current/power monitor [9] is wired in-line on the Pi’s 5 V rail through a 0.1Ω shunt resistor, read over I²C at 1 kHz (verified 99.5 % timing accuracy). A calibration factor of $1.22\times$ corrects a documented under-reading in clone

INA219 modules. Each cryptographic operation is executed 100 times with 50 ms warm-up/cool-down guards; timing, voltage, current, and power arrays are stored in per-operation JSON files for offline analysis.

B. Baseline and Metrics

The idle power with MAVProxy running (no crypto load) is $P_{\text{idle}} = 3,309$ mW. All reported ΔP values are $P_{\text{op}} - P_{\text{idle}}$, isolating the marginal cost of cryptography. Temperature is logged from the CPU thermal zone at 1 Hz.

V. RESULTS

This section reports manually curated benchmark results extracted directly from raw measurement artifacts.

TABLE I
KEM PRIMITIVE PERFORMANCE AND POWER OVERHEAD ON RASPBERRY PI 4 (CORTEX-A72 @ 1.8 GHz, PERFORMANCE GOVERNOR, $n=200$, INA219 @ 1 kHz, $P_{\text{idle}}=3309$ mW).

Algorithm	KeyGen (ms)	Encaps (ms)	Decaps (ms)	ΔP_{kg} (mW)	ΔP_{enc} (mW)	ΔP_{dec} (mW)	PK (B)	SK (B)	CT (B)	SS (B)
<i>NIST Security Level 1</i>										
ML-KEM-512	0.191	0.148	0.153	125.9	153.9	172.7	800	1 632	768	32
HQC-128	22.152	44.625	73.131	412.6	441.6	566.8	2 249	—	4 433	32
Classic-McEliece-348864	297.526	0.460	55.604	966.2	156.2	410.7	261 120	6 492	96	32
<i>NIST Security Level 3</i>										
ML-KEM-768	0.223	0.175	0.181	149.4	185.9	205.7	1 184	2 400	1 088	32
HQC-192	67.430	135.390	211.278	501.2	650.9	731.5	4 522	—	8 978	32
Classic-McEliece-460896	972.453	0.857	89.434	1075.7	279.6	485.1	524 160	13 608	156	32
<i>NIST Security Level 5</i>										
ML-KEM-1024	0.269	0.212	0.225	154.4	180.7	177.5	1 568	3 168	1 568	32
HQC-256	123.836	250.537	392.216	646.9	782.9	819.0	7 245	—	14 421	32
Classic-McEliece-8192128	6901.609	1.856	209.313	1092.6	301.2	639.1	1 357 824	14 120	208	32

TABLE II
DIGITAL SIGNATURE PRIMITIVE PERFORMANCE AND POWER OVERHEAD ON RASPBERRY PI 4 (CORTEX-A72 @ 1.8 GHz, PERFORMANCE GOVERNOR, $n=200$, INA219 @ 1 kHz, $P_{\text{idle}}=3309$ mW).

Algorithm	NIST Level	Latency (ms)			ΔP (mW)			Size (B)		
		KeyGen	Sign	Verify	KeyGen	Sign	Verify	PK	SK	Sig
ML-DSA-44	1	0.368	0.975	0.324	245.8	279.5	235.8	1 312	2 560	2 420
Falcon-512	1	17.722	0.773	0.188	326.9	255.4	238.8	897	1 281	654
SPHINCS ⁺ -SHA2-128s	1	193.609	1 466.336	1.589	729.7	783.7	285.7	32	64	7 856
ML-DSA-65	3	0.551	1.212	0.476	240.9	268.9	240.2	1 952	4 032	3 309
SPHINCS ⁺ -SHA2-192s	3	282.538	2 610.427	2.275	797.1	763.8	303.3	48	96	16 224
ML-DSA-87	5	0.739	1.626	0.718	227.0	268.1	235.2	2 592	4 896	4 627
Falcon-1024	5	49.016	1.441	0.275	439.3	250.4	240.0	1 793	2 305	1 267
SPHINCS ⁺ -SHA2-256s	5	186.472	2 323.760	3.277	715.1	790.5	255.0	64	128	29 792

TABLE III
AEAD CIPHER PERFORMANCE ON MAVLINK PACKET PAYLOADS (RASPBERRY PI 4, CORTEX-A72 @ 1.8 GHz, INA219 POWER MONITORING, $n=10,000$ ITERATIONS PER CONFIGURATION).

Cipher	Payload	Latency (μ s)		Throughput (MB/s)		Power (W)		Energy (μ J/op)		RT (μ J)
		Enc	Dec	Enc	Dec	Enc	Dec	Enc	Dec	
AES-256-GCM	9 B	4.81	4.46	1.87	2.02	3.033	3.025	17.51	12.85	30.36
	54 B	5.79	5.51	9.33	9.81	3.017	3.178	20.38	17.38	37.76
	263 B	9.80	9.59	26.83	27.41	3.078	2.983	32.81	28.65	61.45
ChaCha20-Poly1305	9 B	5.17	4.99	1.74	1.80	3.036	2.992	18.27	13.63	31.91
	54 B	5.25	5.05	10.29	10.70	3.300	2.993	19.99	14.17	34.16
	263 B	6.54	6.49	40.24	40.53	3.037	3.036	22.35	18.60	40.95
Ascon-128a	9 B	11.36	11.16	0.79	0.81	2.970	3.040	49.13	42.41	91.53
	54 B	11.69	11.79	4.62	4.58	3.363	3.041	55.62	44.44	100.06
	263 B	12.87	13.42	20.43	19.59	3.018	3.132	54.28	50.26	104.54

TABLE IV
IMPACT OF DDoS DETECTION ON PQC HANDSHAKE RUNTIME ACROSS NIST SECURITY LEVELS. DATA AGGREGATED FROM 72 HYBRID KEM $\times$ SIG SUITES (SOURCE: `BENCH_DDOS_RESULTS/20260302_135859/COMPARISON.JSON`).

NIST Level	No-DDoS (ms)	XGBoost (ms)	TST (ms)	Δ_{XGB} (ms)	Δ_{TST} (ms)	Suites
L1	699.37	688.86	901.32	−10.51	+201.95	27
L3	1 939.10	1 852.59	2 791.27	−86.51	+852.17	18
L5	6 582.57	4 577.05	6 527.46	−2 005.52 [†]	−55.11	27

[†]Negative Δ at L5 is attributed to model warm-start and OS-level cache effects that offset inference overhead during longer handshakes.

TABLE V
IMPACT OF PQC SECURITY LEVEL ON DDoS DETECTOR RESOURCE COST. AGGREGATED OVER 72 KEM×SIG SUITES (SOURCE: `BENCH_DDOS_RESULTS/20260302_135859/COMPARISON.JSON`).

NIST Level	Suites (n)	CPU (%)					Power (mW)					Throughput (Hz)		
		Absolute			Δ vs. None		Absolute			Δ vs. None		Absolute		
		None	XGB	TST	XGB	TST	None	XGB	TST	XGB	TST	None	XGB	TST
L1	27	21.02	46.21	89.39	+25.19	+68.37	3078.6	4003.9	4107.6	+925.3	+1029.0	22.043	21.464	8.227
L3	18	22.87	48.10	89.64	+25.24	+66.77	3038.8	3996.4	3983.9	+957.6	+945.1	13.406	13.274	5.721
L5	27	22.71	48.16	89.29	+25.45	+66.58	3083.9	4019.4	4113.6	+935.5	+1029.7	17.356	17.191	6.356

TABLE VI
MDEAS SCHEDULER SAFETY THRESHOLDS, DETECTOR OVERHEAD CONSTANTS, AND CROSS-AXIS CONSTRAINTS (SOURCE: `SSCHEDULER/POLICY.PY`).

Panel A: Safety Thresholds			Panel B: Timing & Rekey Limits		
Category	Parameter	Value	Category	Parameter	Value
Battery	Warn	15 200 mV	Rekey	Min stable	60.0 s
	Low	14 800 mV		Max per window	5
	Critical	14 000 mV		Window	300 s
	Rate-warn	500 mV/min		Blacklist TTL	1 800 s
Thermal	Warn	70.0 °C	Hysteresis	Downgrade	5.0 s
	Critical	80.0 °C		Upgrade	30.0 s
	Rate-warn	5.0 °C/min		AEAD recovery	10.0 s
Link Quality	Min PPS	5.0 pkt/s			
	Max gap	1 000 ms			
	Max blackouts	3			

Panel C: Detector Overhead Constants					
Detector	ΔP (W)	ΔT (°C)	ΔCPU (pp)	Warmup (s)	Max T_{base} (°C)
NONE	0.00	0.0	0	0.0	80.0
XGBOOST	0.95	4.8	35	10.0	75.0
TST	1.97	10.7	91	5.0	69.0

Panel D: Cross-Axis Guards

Constraint	Affected Primitives
Forbidden AEAD w/ TST	{Ascon-128a}
Discouraged KEM w/ TST	{Classic-McEliece-348864, Classic-McEliece-460896, Classic-McEliece-8192128}
Forbidden KEM + SIG pairs	{(Classic-McEliece-348864, SPHINCS ⁺ -128s), (Classic-McEliece-460896, SPHINCS ⁺ -192s), (Classic-McEliece-8192128, SPHINCS ⁺ -256s)} + cross-level variants (9 total)

TABLE VII
MDEAS GRACEFUL-DEGRADATION CASCADE: PRIORITY-ORDERED DECISION RULES EXECUTED BY ENERGYAWAREPOLICY (SOURCE: SSCHEDULER/POLICY.PY).

Pri.	Trigger Condition	Action	Threshold	Effect
1	Telemetry stale	HOLD	No sensor data	Freeze all decisions until telemetry resumes
2	DDoS detected	EMERGENCY	DDoS flag raised	Switch to cheapest AEAD (chacha20poly1305), lock exploration
3	Battery critical	EMERGENCY	$V_{\text{bat}} < 14\,000\text{ mV}$	Force L1 (ML-KEM-512 + ML-DSA-44) + cheapest AEAD
4	Temperature critical	EMERGENCY	$T > 80\text{ }^{\circ}\text{C}$	Force L1 + cheapest AEAD
5	Predictive thermal	SWITCH_AEAD	$\hat{T} > \text{thermal budget}$	Proactive AEAD downgrade via ΔT model
6	Thermal / CPU stress	SWITCH_AEAD	Above warn thresholds	Reactive downgrade with break-even check ($\geq 30\text{ s ROI}$)
7	Battery drain stress	SWITCH_AEAD	Rate $> 500\text{ mV min}^{-1}$	Downgrade AEAD with break-even check
8	Link degraded	DOWNGRADE_LEVEL	$< 5\text{ pkt s}^{-1}$ or gap $> 1\,000\text{ ms}$	Reduce NIST security level
9	Cooldown active	HOLD	Hysteresis timer running	Respect timing guard; suppress oscillation
10	AEAD recovery	SWITCH_AEAD	Stress gone $> 60\text{ s}$	Upgrade to preferred AEAD
11	Proactive rekey	REKEY	Timer-based interval	Rotate session keys; same suite ($\approx 300\text{ ms cost}$)
12	Stable + safe	UPGRADE_LEVEL	All clear, 30 s hysteresis	Increase NIST security level (conservative)

Internal constants: DEFAULT_REKEY_COST_MS = 300.0 ms; MIN_BREAK_EVEN_S = 30.0 s; MIN_RECOVERY_STABILITY_S = 60.0 s; ASCON_COST_CEILING = 10

Three-axis model: AEAD: chacha20poly1305 > aesgcm > ascon128a; Security: L1 (ML-KEM-512 + ML-DSA-44) / L3 (768 + 65) / L5 (1024 + 87); DDoS: None / XGBoost

VI. RELATED WORK

Post-quantum security for UAVs has attracted growing attention. Aissaoui [10] benchmarked KEM and digital-signature schemes on a Raspberry Pi for civil UAV communications, reporting execution times but no power or energy measurements. Pham and Vakilinia [11] analysed PQC overhead for UAV communication systems, likewise without energy instru-

mentation. The pqm4 framework [12] provides cycle-accurate benchmarks of ML-KEM and ML-DSA on Cortex-M4 microcontrollers; our Cortex-A72 measurements extend this to companion-class computers with component-level power monitoring.

Rosenpass [13] integrates ML-KEM into WireGuard for stateless post-quantum key exchange, and OQS-OpenSSL [14] enables post-quantum TLS for server environments. Neither system considers energy-constrained scheduling or runtime suite adaptation on embedded platforms.

Prior work on MAVLink security includes proposed protocol-level encryption extensions [1] and surveys of drone communication threats [15], both of which assume classical cryptography and do not address the quantum threat horizon. Cross-platform PQC benchmarks [16] measured algorithm performance across architectures but without energy instrumentation or scheduling logic. Our work is, to our knowledge, the first to combine post-quantum cryptographic deployment with component-level power measurement and runtime adaptive scheduling on an embedded UAV platform.

VII. CONCLUSION

We have presented PQ-TUNNEL, a transparent post-quantum proxy for MAVLink-based UAV communication, and MDEAS, a measurement-driven scheduler that adapts cipher-suite selection at runtime to the aircraft's energy, thermal, and link-quality state. Our INA219-based benchmarks show that lattice-based PQC (ML-KEM, ML-DSA, Falcon) is operationally feasible on Cortex-A72 hardware, with handshakes under 3.5 ms and energy costs below 13 mJ—negligible against the flight battery. The dominant overhead comes not from cryptography but from on-board DDoS detection, whose 1–2 W power draw and 5–11 °C thermal impact require explicit co-scheduling.

MDEAS absorbs these combined overheads through a 15-step policy cascade that prioritises flight safety, preserves encrypted communication, and sheds ancillary services gracefully. The system demonstrates that post-quantum security is not a binary choice for resource-constrained UAVs but a continuous, adaptive capability that a well-instrumented scheduler can manage throughout the mission.

Future work will extend evaluation to outdoor flight with GPS-timestamped telemetry, integrate MDEAS into ArduPilot's companion link manager, and validate the degradation policy under electronic-warfare scenarios combining DDoS and RF jamming.

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